

Functions of Several Variables

$\mathbb{R}$: The set of Real Numbers.

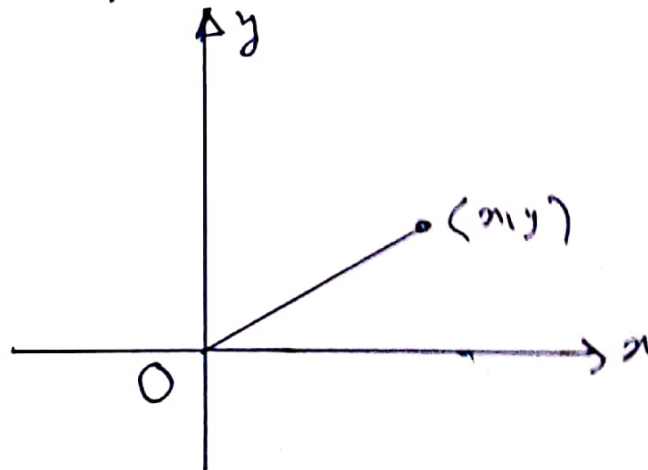


$$\mathbb{R}^n = \left\{ (x_1, x_2, x_3, \dots, x_n) \mid x_i \in \mathbb{R} \text{ for } i = 1, 2, \dots, n \right\}.$$

A typical element $x = (x_1, x_2, \dots, x_n)$ in $\mathbb{R}^n$ is called an ordered n -tuple.

When $n = 2$

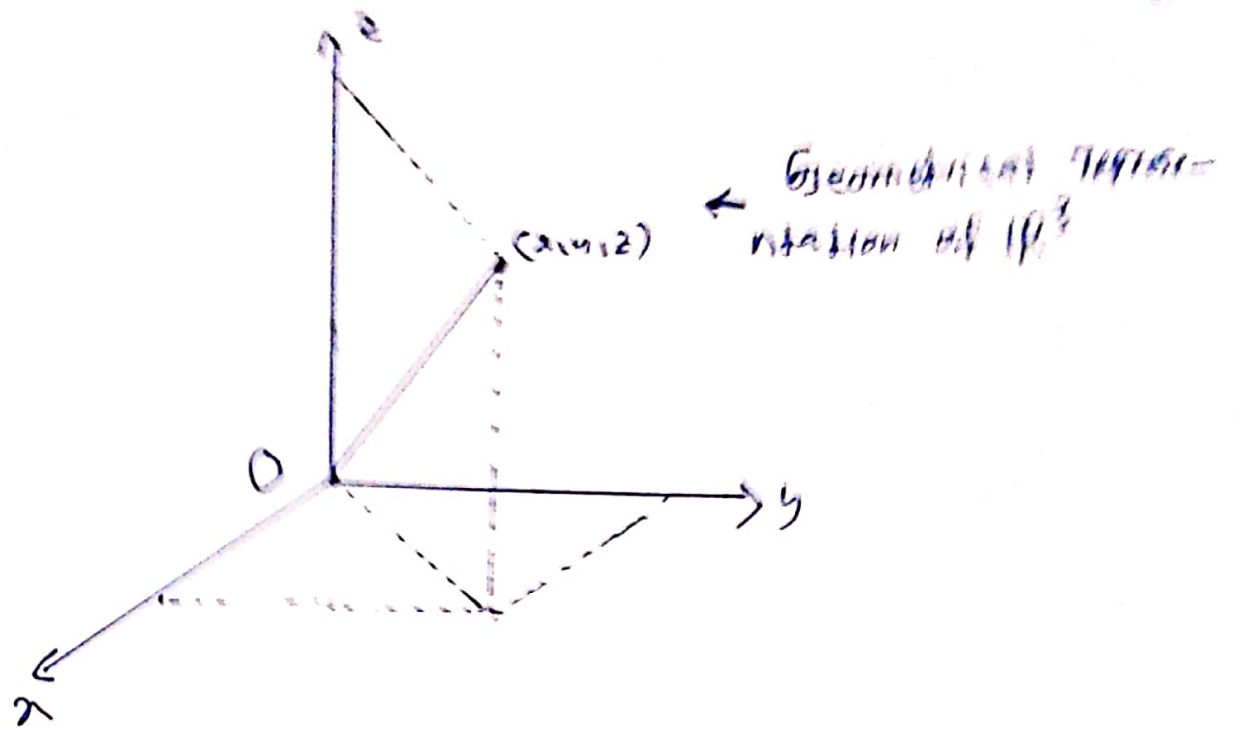
$$\mathbb{R}^2 = \left\{ (x_1, x_2) \mid x_1, x_2 \in \mathbb{R} \right\} = \left\{ (x, y) \mid x, y \in \mathbb{R} \right\}$$



← Geometrical representation of $\mathbb{R}^2$.

When $n = 3$

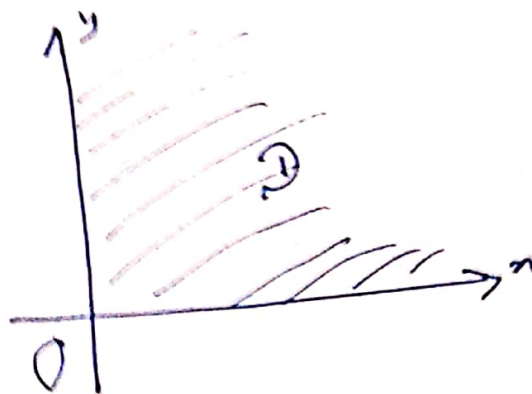
$$\begin{aligned} \mathbb{R}^3 &= \left\{ (x_1, x_2, x_3) \mid x_1, x_2, x_3 \in \mathbb{R} \right\} \\ &= \left\{ (x, y, z) \mid x, y, z \in \mathbb{R} \right\} \end{aligned}$$



In this chapter we shall study functions $f: D \rightarrow \mathbb{R}$, where $D \subseteq \mathbb{R}^2$ or $D \subseteq \mathbb{R}^3$.

Example 1

Let $D = \{(x, y) \mid x > 0, y > 0\}$.



Define $f: D \rightarrow \mathbb{R}$ by $f(x, y) = xy$.

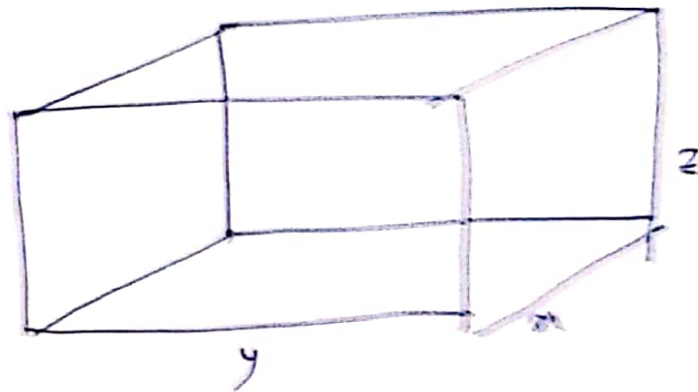
This gives the area of the unbounded region D .

Example 2.

Let $D = \{ (x, y, z) \mid x > 0, y > 0, z > 0 \}$.

Define $V: D \rightarrow \mathbb{R}$ by $V(x, y, z) = xyz$.

This gives the volume of the unbounded cube.



The limit of a function

Note that:

Let $f: \mathbb{R} \rightarrow \mathbb{R}$.

Then $\lim_{x \rightarrow a} f(x) = l \in \mathbb{R}$ if for each $\epsilon > 0$ there

exists $\delta(\epsilon) > 0$ such that

$$|f(x) - l| < \epsilon \text{ whenever } 0 < |x - a| < \delta(\epsilon).$$

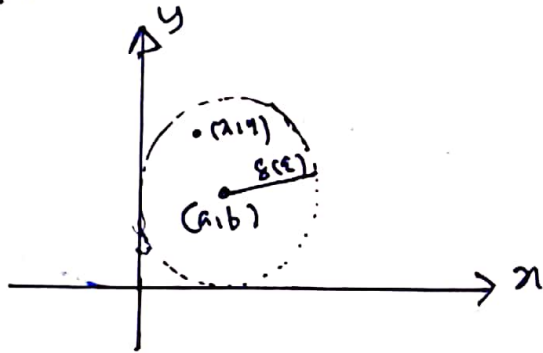
i.e., $f(x)$ can be made arbitrarily close to l by making x sufficiently close to a but not $x = a$.

Definition:

Let $D \subseteq \mathbb{R}^2$ and $f: D \rightarrow \mathbb{R}$.

Then, $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = l \in \mathbb{R}$ if for each $\epsilon > 0$, there exists $\delta(\epsilon) > 0$ such that

$$|f(x,y) - l| < \epsilon \text{ whenever } 0 < \sqrt{(x-a)^2 + (y-b)^2} < \delta(\epsilon).$$



Example 3: Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2 + y^2} = 0$.

Let $f(x,y) = \frac{x^2 y}{x^2 + y^2}$ and let $l = 0$.

The domain of $f = D(f) = \left\{ (x,y) / (x,y) \neq (0,0) \right\} = \mathbb{R}^2 \setminus \{(0,0)\}$

Take any $\epsilon > 0$.

Consider $|f(x,y) - l|$.

$$|f(x,y) - l| = \left| \frac{x^2 y}{x^2 + y^2} - 0 \right| = \left| \frac{x^2 y}{x^2 + y^2} \right| = \frac{x^2}{x^2 + y^2} |y|.$$

$$\leq |y| \quad \text{Since} \quad \frac{x^2}{x^2+y^2} \leq 1 \quad \text{for} \quad (x,y) \neq (0,0).$$

$$\leq \sqrt{x^2+y^2}$$

$$< \epsilon \quad \text{whenever} \quad \sqrt{x^2+y^2} < \epsilon.$$

$$y^2 \leq x^2+y^2$$

$$\Rightarrow |y|^2 \leq x^2+y^2$$

$$\Rightarrow \sqrt{|y|^2} \leq \sqrt{x^2+y^2}$$

$$\Rightarrow |y| \leq \sqrt{x^2+y^2}$$

i.e., $|f(x,y)-0| < \epsilon$ whenever

$$0 < \sqrt{(x-0)^2+(y-0)^2} < \epsilon.$$

We may take $\delta(\epsilon) = \epsilon$.

$$\therefore |f(x,y)-0| < \epsilon \quad \text{whenever} \quad 0 < \sqrt{(x-0)^2+(y-0)^2} < \delta(\epsilon).$$

Then by definition, we get

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y}{x^2+y^2} = 0.$$

H.W.

Prove that $\lim_{(x,y) \rightarrow (0,0)} \frac{3xy^2}{x^2+y^2} = 0.$

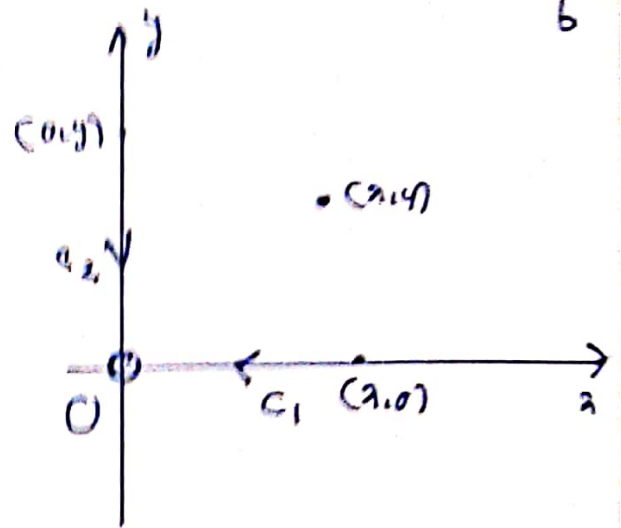
Example 4: Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2-y^2}{x^2+y^2}$ does not exist.

$$D(f) = \mathbb{R}^2 - \{0,0\}$$

Along c_1 : $x > 0$ and $y = 0$.

$$f(x,y) = \frac{x^2 - 0^2}{x^2 + 0^2} = 1$$

$$\lim_{(x,y) \xrightarrow{c_1} (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} 1 = 1 \quad \text{--- (1)}$$



Along c_2 : $y > 0$ and $x = 0$.

$$f(x,y) = \frac{0^2 - y^2}{0^2 + y^2} = -\frac{y^2}{y^2} = -1.$$

$$\therefore \lim_{(x,y) \xrightarrow{c_2} (0,0)} f(x,y) = \lim_{(x,y) \rightarrow (0,0)} -1 = -1 \quad \text{--- (2)}$$

$\therefore$ By (1) and (2), we conclude that

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) \text{ DNE.}$$

Example 5: Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} \text{ DNE.}$

$$\text{Let } f(x,y) = \frac{xy}{x^2 + y^2}.$$

$$D(f) = \mathbb{R}^2 - \{(0,0)\}$$

Along c_1 : $y=0$ and $x>0$.

$$f(x,y) = \frac{x \cdot 0}{x^2 + 0^2} = 0 \text{ Since } x>0$$

$$\therefore \lim_{(x,y) \xrightarrow{c_1} (0,0)} f(x,y) = \lim_{x \rightarrow 0} 0 = 0 \text{ --- (1)}$$

Note that along c_3 : $x=0$ and $y>0$, you will get the same limit 0 and so, we don't consider c_3 .

Along c_2 : $y=x$ and $x>0$.

$$f(x,y) = \frac{x \cdot x}{x^2 + x^2} = \frac{x^2}{2x^2} = \frac{1}{2} \text{ Since } x>0$$

$$\therefore \lim_{(x,y) \xrightarrow{c_2} (0,0)} f(x,y) = \lim_{x \rightarrow 0} \frac{1}{2} = \frac{1}{2} \text{ --- (2)}$$

Hence, by (1) and (2), we conclude that the limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2} \text{ DNE.}$$

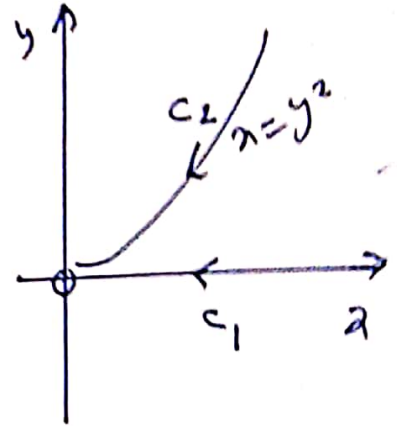
~~limits~~

Example 6

Show that $\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^4}$ DNE.

Let $f(x,y) = \frac{xy^2}{x^2+y^4}$.

Then $D(f) = \mathbb{R}^2 - \{(0,0)\}$



Along c_1 : $x > 0$ and $y = 0$.

$$f(x,y) = \frac{x \cdot 0^2}{x^2 + 0^4} = 0 \quad \text{--- (1)}$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{x \rightarrow 0} 0 = 0 \quad \text{--- (1)}$$

Along c_2 : $x = y^2$ and $y > 0$.

$$f(x,y) = \frac{y^2 \cdot y^2}{(y^2)^2 + y^4} = \frac{1}{2} \quad \text{Since } y > 0.$$

$$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{y \rightarrow 0} \frac{1}{2} = \frac{1}{2} \quad \text{--- (2)}$$

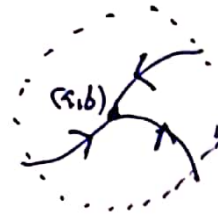
Thus, by (1) and (2), we conclude that the limit

$$\lim_{(x,y) \rightarrow (0,0)} \frac{xy^2}{x^2+y^4} \text{ DNE.}$$

Note:

The definition of $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = L$ refers only to the distance between (x,y) and (a,b) . It does not refer to the direction of approach.

If $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ exists, no matter how (x,y) approaches (a,b) .



Definition:

Let $D \subseteq \mathbb{R}^2$ and $f: D \rightarrow \mathbb{R}$.

f is said to be continuous at $(a,b) \in D$ if

$$\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b).$$

i.e.,

- i) $f(a,b)$ must be defined
- ii) $\lim_{(x,y) \rightarrow (a,b)} f(x,y)$ exists
- iii) $\lim_{(x,y) \rightarrow (a,b)} f(x,y) = f(a,b)$.

Example 7

a) Let $f(x,y) = \frac{x^2y}{x^2+y^2}$

Is f cts at $(0,0)$?

No! Since f is not defined at $(0,0)$.

b) Let $f(x,y) = \begin{cases} \frac{x^2y}{x^2+y^2} & \text{if } (x,y) \neq (0,0) \\ 3 & \text{if } (x,y) = (0,0). \end{cases}$

Is f cts at $(0,0)$?

We have already seen that $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0$.

But $f(0,0) = 3$.

$\therefore \lim_{(x,y) \rightarrow (0,0)} f(x,y) \neq f(0,0)$.

$\therefore f$ is not cts at $(0,0)$.

Definition

Let $D \subseteq \mathbb{R}^2$ and $f: D \rightarrow \mathbb{R}$ be a function. f is said to be continuous on D if f is continuous at each point in D .

Example 8

$$\text{Let } h(x, y) = \begin{cases} \frac{x^2 y}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Since $\lim_{(x, y) \rightarrow (0, 0)} h(x, y) = 0 = h(0, 0)$, f is cts at $(0, 0)$.

In fact, f is cts on $\mathbb{R}^2$.

Partial Differentiation

⑧ Let $f(x, y) = x^2 + y^2$

$$\frac{\partial f}{\partial x} = 2x \leftarrow \text{Regarding } y \text{ as a constant and differentiating } f(x, y) \text{ w.r.t. } x.$$

$$\frac{\partial f}{\partial y} = 2y \leftarrow \text{Regarding } x \text{ as a constant and differentiating } f(x, y) \text{ w.r.t. } y.$$

⑨ Let $f(x, y, z) = xy + yz + z^2$

$$\frac{\partial f}{\partial x} = y \leftarrow \text{Regarding both } y \text{ and } z \text{ as constants and differentiating w.r.t. } x.$$

$$\frac{\partial f}{\partial y} = x + z \leftarrow \text{Regarding both } x \text{ and } z \text{ as constants and differentiating } f(x, y, z) \text{ w.r.t. } y.$$

$$\frac{\partial f}{\partial z} = y + 2z \leftarrow \text{Regarding both } x \text{ and } y \text{ as constants and differentiating } f(x, y, z) \text{ w.r.t. } z.$$

⊗ Let $f(x, y) = e^{x^2+y^2} \sin(x+2y).$

$$\begin{aligned} \frac{\partial f}{\partial x} &= e^{x^2+y^2} \cdot (2x) \sin(x+2y) + e^{x^2+y^2} \cos(x+2y) \\ &= e^{x^2+y^2} [2x \sin(x+2y) + \cos(x+2y)] \end{aligned}$$

$$\begin{aligned} \frac{\partial f}{\partial y} &= e^{x^2+y^2} (2y) \sin(x+2y) + e^{x^2+y^2} \cos(x+2y) (2) \\ &= 2e^{x^2+y^2} [y \sin(x+2y) + \cos(x+2y)] \end{aligned}$$

⊗ Let $f(x, y) = x^3 + 3x^2y + xy^3.$

$$\Rightarrow \frac{\partial f}{\partial x} = 3x^2 + 6xy + y^3$$

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (3x^2 + 6xy + y^3) \\ &= 6x + 6y = 6(x+y). \end{aligned}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (3x^2 + 6xy + y^7)$$

$$= 6x + 7y^6$$

Definition

Let $\mathcal{D} \subset \mathbb{R}^n$ be an open set and let $\underline{a} = (a_1, a_2, \dots, a_n)$ be a point in $\mathcal{D}$.

Let $f: \mathcal{D} \rightarrow \mathbb{R}$.

The partial derivative of f with respect to the i^{th} co-ordinate at $\underline{a}$ is defined to be

$$\lim_{h \rightarrow 0} \left[\frac{f(a_1, a_2, \dots, a_i + h, a_{i+1}, \dots, a_n) - f(a_1, a_2, \dots, a_i, a_{i+1}, \dots, a_n)}{h} \right]$$

$$\lim_{h \rightarrow 0} \left[\frac{f(a_1, a_2, \dots, a_i + h, a_{i+1}, \dots, a_n) - f(a_1, a_2, \dots, a_i, a_{i+1}, \dots, a_n)}{h} \right]$$

provided that the limit exists. This is denoted by $\frac{\partial f(\underline{a})}{\partial x_i}$.

Other notation

- i) $\mathcal{D}_i(\underline{a})$
- ii) $f_{x_i}(\underline{a})$

When $n=1$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$



When $n=2$

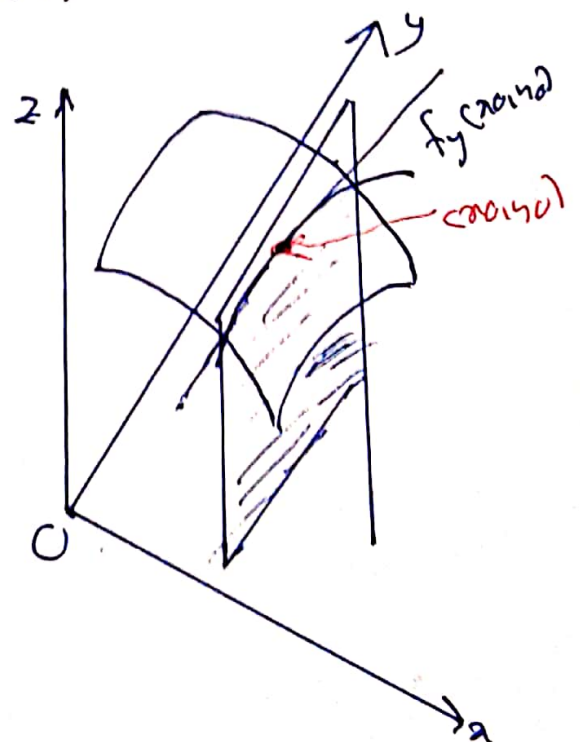
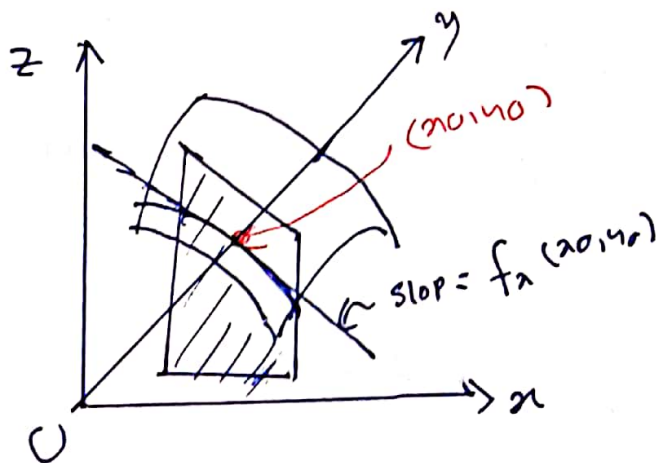
Let $\underline{a} = (x_0, y_0)$.

$$\frac{\partial f(\underline{a})}{\partial x} = \lim_{h \rightarrow 0} \frac{f(x_0+h, y_0) - f(x_0, y_0)}{h}$$

$$= D_1(\underline{a}) = f_x(x_0, y_0).$$

$$\frac{\partial f(\underline{a})}{\partial y} = \lim_{k \rightarrow 0} \frac{f(x_0, y_0+k) - f(x_0, y_0)}{k}$$

$$= D_2(\underline{a}) = f_y(x_0, y_0).$$



Note:

Let $z = f(x, y)$.

Let Δx and Δy be increments w.r.t. x and y , respectively.

Then $\Delta z = f(x + \Delta x, y + \Delta y) - f(x, y)$.

The differential of $f(x, y)$ denoted by $df(x, y)$ and is defined by

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \leftarrow \text{Total differential of } f.$$

Example:

$$\text{Let } f = \begin{cases} \frac{xy}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

We have already seen that f is not continuous at $(0, 0)$.

When $(x, y) \neq (0, 0)$ $f = f(x, y)$ is cts.

$$f(x, y) = \frac{xy}{x^2 + y^2}.$$

$$\frac{\partial f}{\partial x} = \frac{(x^2 + y^2)y - xy(2x)}{(x^2 + y^2)^2} = \frac{y^3 - x^2y}{(x^2 + y^2)^2}$$

Similarly,

$$\frac{\partial f}{\partial y} = \frac{x(y^2 - x^2)}{x^2 + y^2}$$

$$= \frac{y(y^2 - x^2)}{(x^2 + y^2)^2}$$

Do $\frac{\partial f}{\partial x}(0,0)$ and $\frac{\partial f}{\partial y}(0,0)$ exist?

By definition,

$$f_x(0,0) = \frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(0+h,0) - f(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{0-0}{h}$$

$$= 0.$$

$$f_y(0,0) = \frac{\partial f}{\partial y}(0,0) = \lim_{k \rightarrow 0} \frac{f(0,0+k) - f(0,0)}{k}$$

$$= \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k}$$

$$\frac{\partial f}{\partial x} = \frac{(x^2 + y^2)y - xy(2x)}{(x^2 + y^2)^2} = \frac{y^3 - x^2y}{(x^2 + y^2)^2}$$

Similarly,

$$\frac{\partial f}{\partial y} = \frac{x(y^2 - x^2)}{x^2 + y^2}$$

$$= \frac{y(y^2 - x^2)}{(x^2 + y^2)^2}$$

Do $\frac{\partial f}{\partial x}(0,0)$ and $\frac{\partial f}{\partial y}(0,0)$ exist?

By definition,

$$f_x(0,0) = \frac{\partial f}{\partial x}(0,0) = \lim_{h \rightarrow 0} \frac{f(0+h,0) - f(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h,0) - f(0,0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{0-0}{h}$$

$$= 0.$$

$$f_y(0,0) = \frac{\partial f}{\partial y}(0,0) = \lim_{k \rightarrow 0} \frac{f(0,0+k) - f(0,0)}{k}$$

$$= \lim_{k \rightarrow 0} \frac{f(0,k) - f(0,0)}{k}$$

$$= \lim_{k \rightarrow 0} \frac{0-0}{k}$$

$$= 0.$$

Both $f_x(0,0)$ and $f_y(0,0)$ exist at $(0,0)$.

But f is not continuous at $(0,0)$.

i.e., the existence of the first partial derivatives at a point does not imply continuity at a point.

Differentiability of functions of two variables

Let f be a function defined in some neighbourhood of (a,b) .

Suppose that $f_x(a,b)$ and $f_y(a,b)$ exist.

Then f is said to be differentiable at (a,b) if

$$\lim_{(h,k) \rightarrow (0,0)} \left[\frac{f(a+h, b+k) - f(a,b) - hf_x(a,b) - kf_y(a,b)}{\sqrt{h^2 + k^2}} \right] = 0.$$

Example:

$$\text{Let } f(x, y) = 2x^2y.$$

Use the definition to show that f is differentiable at $(2, 3)$.

Let $a=2$ and $b=3$.

$$\text{Let } R(h, k) = \frac{f(a+h, b+k) - f(a, b) - hf_x(a, b) - kf_y(a, b)}{\sqrt{h^2 + k^2}} \quad \text{--- (1)}$$

We must show that $\lim_{(h, k) \rightarrow (0, 0)} R(h, k) = 0$.

$$f(x, y) = 2x^2y \Rightarrow f_x(x, y) = 4xy, \quad f_y(x, y) = 2x^2.$$

$$\text{Hence, } f_x(2, 3) = 24 \text{ and } f_y(2, 3) = 8.$$

$$\text{Also, } f(2, 3) = 24 \text{ and } f(2+h, 3+k) = 2(2+h)^2(3+k).$$

Now by (1), we get

$$R(h, k) = \frac{2(2+h)^2(3+k) - 24 - h \times 24 - 8k}{\sqrt{h^2 + k^2}}$$

$$= \frac{8hk + 2h^2k + 6h^2}{\sqrt{h^2 + k^2}}$$

$$= \frac{8hk}{\sqrt{h^2 + k^2}} + 2 \frac{h^2k}{\sqrt{h^2 + k^2}} + 6 \frac{h^2}{\sqrt{h^2 + k^2}}$$

$$\lim_{(h,k) \rightarrow (0,0)} R(h,k) = 8 \lim_{(h,k) \rightarrow (0,0)} \frac{hk}{\sqrt{h^2+k^2}} + 2 \lim_{(h,k) \rightarrow (0,0)} \frac{h^2k}{\sqrt{h^2+k^2}} + 6 \lim_{(h,k) \rightarrow (0,0)} \frac{h^2}{\sqrt{h^2+k^2}} \quad (2)$$

Take any $\epsilon > 0$.

$$\text{Now, } \left| \frac{hk}{\sqrt{h^2+k^2}} - 0 \right| = \left| \frac{hk}{\sqrt{h^2+k^2}} \right| = |h| \cdot \frac{|k|}{\sqrt{h^2+k^2}}$$

$$\leq |h|$$

$$\leq \sqrt{h^2+k^2}$$

$$\leq \epsilon \text{ whenever } 0 < \sqrt{h^2+k^2} < \epsilon$$

$$\therefore \lim_{(h,k) \rightarrow (0,0)} \frac{hk}{\sqrt{h^2+k^2}} = 0.$$

$$\leq \epsilon \text{ whenever } 0 < \sqrt{h^2+k^2} < \delta(\epsilon)$$

Similarly, we can show that

$$\lim_{(h,k) \rightarrow (0,0)} \frac{h^2k}{\sqrt{h^2+k^2}} = 0 \text{ and } \lim_{(h,k) \rightarrow (0,0)} \frac{h^2}{\sqrt{h^2+k^2}} = 0.$$

Now by (2), we get $\lim_{(h,k) \rightarrow (0,0)} R(h,k) = 0$.

$\therefore f$ is differentiable at $(2,3)$.

Note that f is differentiable at (a,b)

$$\Leftrightarrow \lim_{(h,k) \rightarrow (0,0)} R(h,k) = 0.$$

Theorem

If a function $f = f(x, y)$ is differentiable at (a, b) , then f is continuous at (a, b) .

— Second order derivatives —

Let $f = f(x, y)$.

Suppose that $f_x(x, y)$ and $f_y(x, y)$ exist.

$$f_x = \frac{\partial f}{\partial x} \Rightarrow \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial x} (f_x) = f_{xx}$$

$$f_y = \frac{\partial f}{\partial y} \Rightarrow \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial y} (f_y) = f_{yy}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} (f_y) = f_{xy}$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} (f_x) = f_{yx}$$

Note that, in general, $f_{xy} \neq f_{yx}$.

H.W

Find an example for a function $f = f(x, y)$ such that $\frac{\partial^2 f}{\partial x \partial y} \neq \frac{\partial^2 f}{\partial y \partial x}$.

Example

$$\text{Let } f(x, y) = x^3 y + e^{xy^2}$$

$$f_x = \frac{\partial f}{\partial x} = 3x^2 y + e^{xy^2} \cdot y^2$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} (f_x) = 3x^2 + 2y e^{xy^2} (xy^2 + 1) \quad \text{--- (1)}$$

$$f_y = \frac{\partial f}{\partial y} = x^3 + 2xy e^{xy^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} (f_y)$$

$$= 3x^2 + 2y [e^{xy^2} + xy^2 e^{xy^2}]$$

$$= 3x^2 + 2y e^{xy^2} (1 + xy^2) \quad \text{--- (2)}$$

Then it is clear that

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

Theorem:

If f_x and f_y are both differentiable at a point (a, b) of the domain of a function f , then $f_{xy}(a, b) = f_{yx}(a, b)$.

Theorem

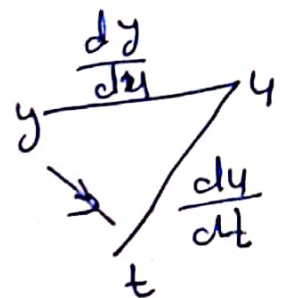
If f_y exists in a neighbourhood of a point (a, b) of the domain of the definition of a function f and f_{yx} is continuous at (a, b) , then $f_{xy}(a, b)$ exists and

$$f_{xy}(a, b) = f_{yx}(a, b).$$

Chain Rule.

If $y = f(u)$ is a differentiable function of u and $u = g(t)$, then

$$\frac{dy}{dt} = \frac{dy}{du} \cdot \frac{du}{dt}$$



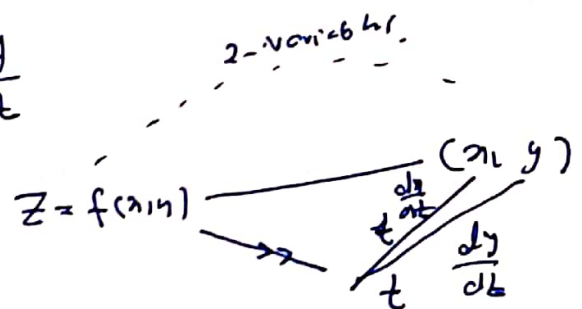
Suppose that $x = g(t)$ and $y = h(t)$ are both differentiable functions of t .

Let $z = f(x, y)$ be a differentiable function of x and y .

Suppose that $x = g(t)$ and $y = h(t)$ are both differentiable functions of t .

Then z is a differentiable function of t and

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt}$$



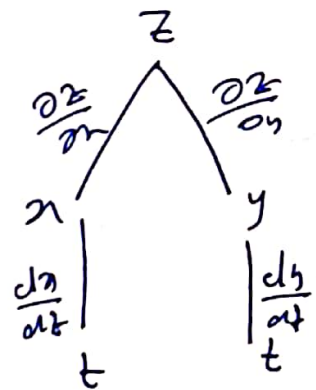
Example

Let $z = e^x \sin y$, $x = t^2$ and $y = \sin t$.

Find $\frac{dz}{dt}$

By chain rule,

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \frac{dx}{dt} + \frac{\partial z}{\partial y} \frac{dy}{dt} \quad \text{--- (1)}$$



$$z = e^x \sin y.$$

$$\Rightarrow \frac{\partial z}{\partial x} = e^x \sin y \quad ; \quad \frac{\partial z}{\partial y} = e^x \cos y.$$

$$x = t^2 \Rightarrow \frac{dx}{dt} = 2t$$

$$y = \sin t \Rightarrow \frac{dy}{dt} = \cos t.$$

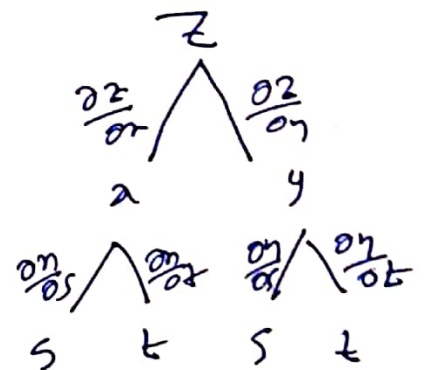
Now by ① we get

$$\begin{aligned}\frac{dz}{dt} &= e^x \sin y \cdot 2t + e^x \cos y \cdot \cos t \\ &= e^x (2t \sin y + \cos t \cos y).\end{aligned}$$

Let $z = f(x, y)$ be a differentiable function of x and y .

Suppose that $x = g(s, t)$ and $y = h(s, t)$, where g_s, g_t, h_s , and h_t all exist.

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$



$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

Example.

Let $z = e^x \sin y$, $x = s \cdot t^2$, $y = s^3 \cdot t$.

Find $\frac{\partial z}{\partial s}$ and $\frac{\partial z}{\partial t}$.

$$\begin{aligned}\frac{\partial z}{\partial s} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s} \\ &= (e^x \sin y)(t^2) + (e^x \cos y)(3s^2 t) \\ &= e^x t (t \sin y + 3 \cos y \cdot s^2)\end{aligned}$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \frac{\partial y}{\partial t}$$

$$z = e^x \sin y.$$

$$= (e^x \sin y)(2ts) + (e^x \cos y) \cdot s^3$$

$$= e^x (2ts \sin y + s^3 \cos y).$$

Examples:

H.W

Let $z = f(x, y)$, $x = r^2 + s^2$ and $y = 2rs$.

Find $\frac{\partial z}{\partial r}$, $\frac{\partial z}{\partial s}$, $\frac{\partial^2 z}{\partial s^2}$, $\frac{\partial^2 z}{\partial s \partial r}$ and $\frac{\partial^2 z}{\partial r^2}$

assuming that $z_{xy} = z_{yx}$.